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Dependency Distance and Its Probability Distribution: Are They the Universals for Measuring Second Language Learners' Language Proficiency?

Yuxin Hao^a, Xuelin Wang^a and Yanni Lin ^b

^aInstitute of Chinese Language and Culture Education, Huaqiao University, Xiamen, China;

^bCollege of Foreign Studies, Guangxi Normal University, Guilin, China

ABSTRACT

Previous studies have shown that dependency distance and its probability distribution can be applied as syntactic indicators of English as interlanguage. However, the universal application of these indicators has not been verified from the perspective of language typology. The issues are addressed in the present study based on a treebank of Chinese interlanguage of English and Japanese native speakers. The findings are as follows: (1) with the improvement of L2 proficiency, the MDDs of learners with different native language backgrounds gradually approach that of the target language in different patterns, and dependency distance is of universal significance as a metric to measure the development of interlanguage's syntactic complexity; (2) Chinese interlanguage also follows the principle of least effort, and its probability distribution of dependency distance, like those of natural languages, presents a power-law distribution, which can successfully fit the Zipf-Alekseev distribution; (3) the right truncated modified Zipf-Alekseev distribution can be used to measure Chinese interlanguage proficiency, and the fitting parameters of the probability distribution of dependency distance as a metric of interlanguage proficiency are also of universal value.

1. Introduction

Quantitative Linguistics takes language materials produced from real communications as research objects, attempting to describe and analyse various linguistic phenomena, linguistic structures, structural attributes, their relationships and language evolution by means of quantitative methods. 'Precise', 'real' and 'dynamic' are its three main features (Herdan, 1964; Köhler et al., 2005; Liu, 2017; Zipf, 1936). After decades of efforts, quantitative linguists have discovered a number of universal laws. For example, the law of distribution represented by Zipf's law (Zipf, 1936), the law of function represented by Menzerath-Altmann's law (Altmann, 1980) and the law of

evolution represented by Piotrowski-Altmann's law (Beöthy & Altmann, 1982). These laws provide a necessary way and an effective means for the scientific study of language.

Among them, Zipf's law is one of the earliest statistical laws proposed by quantitative linguistics, which is closely related to many language properties and language processes, and has been confirmed in many languages. It has a core position in quantitative linguistics. Popescu et al. (2014) found a variant of Zipf's law, the Zipf-Alekseev model, which proved to fit the length distribution of units at different levels in a language, simplifying the complex problem of the unit length distribution in natural languages. Dependency distance (DD) is one of the lengths between language units, which is the linear distance between two words in a sentence that have a dominant and dominated relationship (Heringer et al., 1980; Hudson, 1995). Previous studies have discovered a strong correlation between dependency distance, human working memory and cognitive load (Gibson, 1998; Hudson, 2007; Morrill, 2000). Dependency distance is often considered a metric of syntactic complexity and language processing difficulty (Gibson, 2000; Gildea & Temperley, 2010; Hawkins, 2004; Temperley, 2007). Based on the mean dependency distance (MDD) of 20 languages, Liu (2008) as well as Liu, Hudson et al. (2009) found that human language has a threshold of mean dependency distance, which is maintained by the cooperation of grammar and cognition. Strauss et al. (2006) stated that the rank law of each dependency distance unit follows a certain rank frequency distribution model.

It was also found that the probability distribution of dependency distance in natural language conforms to a certain distribution law under the conditions of controlled and uncontrolled sentence lengths. The probability distribution of mixing dependency distances from sentences of different lengths in natural languages conforms to the right-truncated Zeta distribution, including the Zipf-Alekseev model (Jiang & Liu, 2015; Liu, 2007), while that of dependency distance of sentences of the same length is a double-exponential (Ferrer-i-Cancho, 2004). In addition, it was found that syntactic complexity in different languages appeared with different measures. For example, Jing and Liu (2015) found that there are differences between English and Czech in the measurement of syntactic complexity and real-time processing difficulty, for example, for longer sentences, English prefers to increase the MDD, while Czech tends to enhance the MHD (mean hierarchical distance).

Interlanguage (hereafter IL) is a concept in the field of second language acquisition, which refers to a natural language system produced by second language learners when they are acquiring a new language (Richards et al., 1996). IL is an independent and progressive dynamic continuous system (Corder, 1967; Nemser, 1971; Selinker, 1972), which shares the

characteristics of both the learner's native language and the target language (Brown, 1994; Selinker, 1972). Based on the dynamic development of interlanguage (Larsen-Freeman, 1997), one of the most important topics in the field of second language acquisition is finding the metrics that can measure the development of second language (hereafter L2) proficiency development. Among them, syntactic complexity, which represents the scope and complexity of language output forms, has attracted considerable attention and is considered a metric to measure the overall levels of L2 proficiency and language development (Lu, 2011; Ortega, 2003). Researchers have long tried to quantify syntactic complexity with various metrics (Wolfe-Quintero et al., 1998). The length of the output unit is one of the quantitative indicators. In view of the natural language properties of interlanguage (Selinker, 1972), the developmental laws of interlanguage can be revealed based on the research achievements of quantitative linguistics. Previous studies have shown that the methods used in quantitative linguistics work (Bi & Jiang, 2020; Chen & Xu, 2019; Eppler, 2011; Jiang & Ouyang, 2017; Ouyang & Jiang, 2018). For example, similar to the results of native natural language research, in the case of mixing dependency distances from sentences of different lengths, Jiang and Ouyang (2017) as well as Li and Yan (2020) have shown that MDD could indicate the syntactic complexity of the learners' English. In other words, closely related to human working memory, dependency distance as the metric of syntactic complexity can also reveal the development of syntactic complexity of interlanguage to a great extent. According to the studies of Jiang and Ouyang (2017) and Li and Yan (2020), foreign English learners' MDDs increase with their ILs' proficiency, which proves that MDD can be used as a metric of the development of interlanguage syntactic complexity. With the mixing of dependency distances from sentences of different lengths, Yan and Li (2019) found the MDD of Chinese L2 writing appears to increase as the proficiency level advances from low to high, and significant differences were found between all the pairs ($ps < 0.05$). In the respect of the probability distribution of interlanguage's dependency distance, studies of Ouyang and Jiang (2018) and Li and Yan (2020) based on syntactically tagged corpora of English interlanguage have proved that English interlanguage, like natural languages, is also constrained by working memory, and the probability distribution of dependency distance fits well with the right truncated modified Zipf-Alekseev distribution. In the Zipf-Alekseev formula, parameter a is interpreted as the feature of the language at large, whereas parameter b shows the changes made by the author of the text (Hřebíček, 2002, cited from; Andreev, 2020). It was also found that the parameters a , b of the right truncated modified Zipf-Alekseev distribution in interlanguage increase and decrease with the increasing of language proficiency, respectively, indicating that parameters of the right truncated

modified Zipf-Alekseev distribution can also be used to measure the language proficiency of second language learners. However, Komori et al. (2019) advocated the MHD might be a better measure to show Japanese learners' syntactic development than MDD for both intermediate and advanced learners. The reason is that there may be some linguistic preferences between the MDD and the MHD in Japanese, and some of the characteristics of Japanese syntactic complexity are better displayed when using MHD rather than MDD (Komori et al., 2019).

Nevertheless, the application of the quantitative linguistics method to second language acquisition is still minority, and there are still some deficiencies and gaps in the above research.

Firstly, the previous studies mainly focus on the development of English (Indo-European language) interlanguage's dependency distance. Whether the previous findings are accidental or not needs further verification in other languages as interlanguage such as Chinese, which is a Sino-Tibetan language. Can dependency distance in Chinese interlanguage measure L2 learners' syntactic development? Does probability distribution of dependency distance in Chinese interlanguage fit well the right truncated modified Zipf-Alekseev distribution? If it is possible, can the parameters in the Zipf-Alekseev distribution measure the language proficiency of Chinese learners? In other words, are dependency distance and its probability distribution universal metrics to measure second language learners' language proficiency? All of these require further exploration.

Secondly, previous studies clearly show that different languages have different dependency distances (Eppler, 2010; Ferrer-i-Cancho et al., 2020; Hiranuma, 1999; Jiang & Liu, 2015; Liu, 2008). According to Liu (2008), Chinese MDD (3.662) is the longest in the 20 languages, while those of English (SVO language) and Japanese (SOV language) are 2.543 and 1.805, respectively. It is believed that the differences in terms of language typology and the language distance between the native language and the target language have an impact on second language acquisition (Gass & Selinker, 1992; Odlin, 1989). However, as far as the interlanguage dependency distance is concerned, few scholars compare the features of learners from different native language backgrounds who acquire the same second language from the perspective of linguistic typology. Does the native language of second language speakers have an impact on the development of interlanguage's MDD? Put differently, what are the similarities and differences of interlanguage's MDD at the initial stage, the development process and the final state when learners with different native language backgrounds are acquiring the same second language? Can they reach the same learning level? It is necessary to study the interlanguage of two types of comparable Chinese Second Language (hereafter CSL) learners with different native language backgrounds simultaneously.

Most of the previous studies focused on English as the interlanguage. To fill the above-mentioned gaps, we will expand the scope of the research sample and choose Chinese learners' writing materials to investigate the development of Chinese interlanguage's MDD. Furthermore, we also examine the validity of the probability distribution of dependency distance as the metrics to measure second language's proficiency. It should be noted that, we chose MDD as the measurement because, as we mentioned above, Chinese and English have similarities regarding linguistic typology, both Chinese and English are SVO languages (Dryer, 2013), and are bidirectional dependency languages as well (Liu, 2010). Characteristics of Chinese syntactic complexity are also likely to appear with MDD rather than MHD. Consequently, the current study follows Jiang and Ouyang (2017), Ouyang and Jiang (2018), further investigating the validity of MDD rather than MHD as a metric of Chinese interlanguage development. This paper involves ECSL learners (English native-speaking Chinese second language learners) and JCSL learners (Japanese native-speaking Chinese second language learners). The first reason of choosing the two kinds is that there are differences in native language backgrounds between two kinds of CSL learners. English (native language, NL) and Chinese (target language, TL) are SVO languages with balanced distribution of dependency directions, while Japanese (NL) is a SOV language, which is predominantly head-final (Liu, 2010). Secondly, there are great differences in MDD between learners with different native language backgrounds (English and Japanese), which is helpful for us to observe the development of their interlanguages more clearly. Therefore, in terms of the linguistic type of learners' NL, the composition data examined in this paper can be divided into two types: one is similar to the TL and the other is far from the TL. Based on the linguistic typology, this paper makes an in-depth study on the similarities and differences of interlanguage's dependency distance and its probability distribution with different native language backgrounds. Specific research questions are as follows:

Question 1: Does IL's MDD of CSL learners with different L1 backgrounds vary and develop as the language proficiency improves? What are the differences and similarities between them?

Question 2: Does the probability distribution of dependency distance of CSL learners' MDD successfully fit the Zipf-Alekseev distribution?

Question 3: If CSL learners' interlanguage follows the power-law distribution, are the parameters of their probability distribution model of dependency distance related to their language proficiency? Can the parameters be used as universal metrics to measure L2 proficiency?

Question 4: Are second language learners’ interlanguage dependency distance and its probability distribution influenced by the correspondence between their native language type and the target language type?

2. Methods and Materials

Within the theoretical framework of dependency grammar, syntactically tagged corpora (treebanks) provide an effective support for the quantitative study of syntax, which makes the syntactic study more precise and accurate. Dependency grammar is a syntactic theory based on the syntactic dependency between the governor and the dependent (Hudson, 2007; Tesnière, 1959). Dependency grammar holds that a sentence is an organized and complete structural unit whose basic components are words. Sentence structures are determined by the dependency relationships between sentence components. The task of syntactic analysis is to determine the dependency relationship between the governor and the dependent (Liu, 2009). The dependency relation is formed between two asymmetrical, directional syntactically related structural elements. One is the governor and the other is the dependent. According to these properties of dependency grammar, we can construct dependency structures marked by directed arcs. Figure 1 shows the dependency analysis of the sentence *Wo Zai Ribendaxue Zhongwenxi Biye* (I graduated from the Chinese Department of Nihon University). The syntactic label above the arc indicates the specific type of dependency relation between the governor and the dependent. Figure 1 shows the dependency structure of the governor and the dependent in the sample sentence, reflecting the linear order of the two components that construct the dependency relationship. Table 2 presents a more computer-friendly format of syntactic analysis results shown in Figure 1. In Table 2, each row has a dependency pair consisting of four main parts, namely the dependent word, the governor word, the dependency type, and the dependency

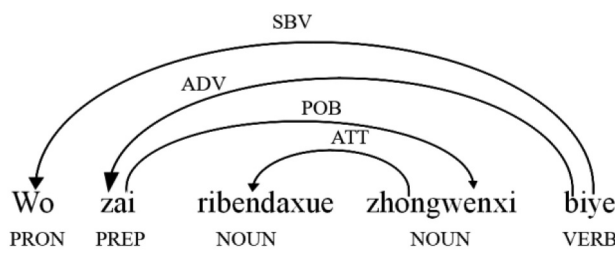


Figure 1. Dependency Structure of sample sentence in Chinese: *Wo zai ribendaxue zhongwenxi biye* (I graduated from the Chinese Department of Nihon University).

distance. The order numbers of the governor and dependent words are clearly marked in the corpus, and the difference between the two numbers represents the linear distance (Dependency Distance) between the governor and dependent words (Liu, Hudson et al., 2009).

2.1. Materials

The Chinese interlanguage examined in this paper comes from the HSK composition Corpus. HSK (from Level I to VI) is an international standardized test with a history of more than 30 years, which assesses non-native Chinese speakers' abilities of using the Chinese language in their daily, academic and professional lives. According to the scoring system of HSK Test, the composition score is not absolutely continuous. Theoretically speaking, it is an arithmetic sequence ranging from 0 to 100 with a common difference of 5. In reality, there are very few compositions with extremely low or high scores. For some practical considerations, we discarded the compositions scored below 40. Specifically, extremely low scores account for only a very small proportion in the HSK composition corpus. In our sample made of 801 compositions, there were only 7 zero-rated ones ($7/801 \approx 0.87\%$), the length of which was very short – with only dozens of words. In addition, the authors of the seven compositions are not native speakers of the English and Japanese, whereas our study focuses on ECSL and JCSL. In the HSK composition Corpus from which 801 compositions were randomly chosen, the scores fall within the range from 40 to 95. Based on the real-world data represented by the corpus, there are altogether 12 types of scores: {40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95}. The length of compositions totaled 186,650 tokens for both English and Japanese native speakers. The compositions of CSL learners with two different NL backgrounds were classified roughly into four levels (P1-P4), each level with approximately 20,000 tokens after removing the punctuation. The results of one-way ANOVA show that the composition scores of both ECSL learners ($F(3,424) = 1476.306$) and JCSL learners ($F(3,464) = 1677.422$) are significantly different among the four grades ($p < .001$). A post-hoc analysis using Tukey-HSD shows that significant differences are found between the composition scores of ECSL and JCSL learners between all pairs ($ps < .001$). And this means that the range of scores can distinguish their Chinese levels. We tried to keep the consistency of composition's genre and topic at each grade. L2 learners are familiar with all the topics of compositions, such as 'Remember a person who has the greatest influence on me', 'The joys and frustrations of learning Chinese', 'A holiday for me' and so on. Two contrastive treebanks of native Chinese were extracted from Chinese modern novels, each consisting of around 20,000 tokens after removing the punctuation. In order to ensure

Table 1. A brief information of each proficiency.

NL	Average score of essays (min, max)	L2 proficiency	Number of essays	Tokens
English	40–60	P1	136	20,066
	65–70	P2	91	20,026
	75–80	P3	81	20,041
	85–95	P4	76	20,003
Japanese	40–60	P1	144	20,056
	65–70	P2	106	20,049
	75–80	P3	87	20,092
	85–95	P4	80	20,079

Note: The number of compositions is equal to the number of individuals within each proficiency level and group.

Table 2. Annotation of the sample sentence in Chinese, *Wo zai ribendaxue zhongwenxi biye* (I graduated from the Chinese Department of Nihon University).

Dependent			Governor			Dependency	DD	ADD
Number	Word	POS	Number	Word	POS			
1	我(Wo/I)	r	5	毕业(biye/graduate)	v	SBV	4	4
2	在(zai/in)	p	5	毕业(biye/graduate)	v	ADV	3	3
3	日本大学(ribendaxue/Nihon University)	n	4	中文系(zhongwenxi/Chinese Department)	n	ATT	1	1
4	中文系(zhongwenxi/Chinese Department)	n	2	在(zai/in)	p	POB	–2	2
5	毕业(biye/graduate)	v	0	/	/	HED	0	0

Note: In Table 2, ‘DD’ is short for Dependency Distance, and ‘ADD’ for Absolute Dependency Distance. All the words in the second column (‘Word’) are originally Chinese characters, each followed by the corresponding Pinyin note and English translation.

representativeness of sampling, the target language’s (Chinese) datum used for comparison took the mean of the test scores. The number of texts and the score ranges of each level are shown in Table 1.

2.2. Procedure

The basic annotation process can be divided into the following procedure, namely, word segmentation, part-of-speech (POS) tagging, dependency syntactic tagging and error tagging. First of all, the part of speech annotation and dependency relation tagging were automatically completed by Language Technology Platform (LTP) (Che et al., 2010), a Chinese processing platform developed by the Research Center for Social Computing and Information Retrieval at Harbin Institute of Technology. Then, the manual proofreading and error labelling were carried out on the basis of preliminary machine labelling. We retained dependency relations of L2 learners’ syntactic errors. In this way, we can ensure not only the efficiency and accuracy of annotation,

but also the reliability of research results of this paper. To facilitate the calculation, we then used Python scripts to convert the information shown in Figure 1 into Table 2.

2.3. Data Analysis

Dependency distance is the difference between the order number of the governor and the order number of the dependent. As the governor may precedes or follows the dependent, the value of dependency distance can be positive or negative, which indicates the dependency direction. However, we need to use the absolute value of DD (ADD) to calculate the mean dependency distance (MDD) of a sentence or the sample. For example, the dependency distance of a sentence containing a number of n words is derived from formula (1):

$$MDD(\text{the sample}) = \frac{1}{n-1} \sum_{i=1}^{n-1} |DD_i| \quad (1)$$

In a sentence, the only word that does not have a governor is the root node of the syntax tree, labelled as HED. The DD of HED is considered as zero. Thus, we can get several DDs for the sample sentence: 4, 3, 1, -2, 0, which are obtained by subtracting the order number of the word and that of its governor.

Then, by Formula (1), the MDD of the sample sentence is $(|4| + |3| + |1| + |-2| + |0|)/4 = 2.5$. If we generalize the formula (1) to a treebank, we can get MDD for the whole treebank. If there are m words and n sentences in a treebank, the MDD of a treebank is:

$$MDD(\text{the sample}) = \frac{1}{m-n} \sum_{i=1}^{m-n} |DD_i| \quad (2)$$

The DD_i in Equations (1) and (2) is the dependency distance of the i -th dependency in a sentence or a treebank.

Zipf-Alekseev model can fit not only the mixing dependency distances from sentences of different lengths of natural languages successfully (Jiang & Liu, 2015; Lu & Liu, 2016), but also English interlanguage with different L1 backgrounds (Li & Yan, 2020; Ouyang & Jiang, 2018). Based on the results of previous studies, we hypothesize that the probability distribution of ECSL learners' and JCSL learners' MDDs of Chinese interlanguages also follow the Zipf-Alekseev model (Hřebíček, 1996; cited from; Strauss & Altmann, 2006). Hřebíček adopted two assumptions:

(1) the logarithm of the ratio of the probabilities P_1 and P_x is proportional to the logarithm of the class size, i.e.

$$\ln(P_1/P_x) \propto \ln x$$

(2) the proportionality function is given by the logarithm of Menzerath's law (Hierarchy), i.e.

$$\ln(P_1/P_x) = \ln(AX^b)\ln x$$

Popescu et al. (2014) have stated that the relative rate of change of the dependent variable (here is the frequency of the dependency distance) is proportional to the rate of change of the independent variable (here is the value of dependency distance). The Zipf-Alekseev distribution can be expressed in formula (3):

$$P_x = P_1 x^{-(a+b\ln x)}, x = 1, 2, 3 \dots \quad (3)$$

P_1 is the logarithm of the ratio of the probabilities and P_x is proportional to the logarithm of the class size. If formula (3) is considered a probability distribution, then P_1 is the normalization constant; otherwise, it is the size of the first category, which is the case of $x = 1$. In most cases, diversification distributions display a diverging frequency in the first class, while the rest of the distributions behave regularly. In such cases, the first class is usually assigned to a particular value, α ; then, formula (3) can be transformed to the following formula:

$$P_x = \begin{cases} \alpha, & x = 1 \\ \frac{(1-\alpha)x^{(a+b\ln x)}}{T}, & x = 2, 3, \dots, (n) \end{cases} \quad (4)$$

where:

$$T = \sum_{j=2}^n j^{-(a+b\ln j)}, 0 < \alpha < 1, a, b \in \mathbb{R}$$

The formula (3) and (4) are called the Zipf-Alekseev distribution. If n is finite, then formula (4) is called the right truncated modified Zipf-Alekseev distribution. In this study, we will use the software, Altmann-Fitter (Altmann-Fitter, 2013), to fit Chinese interlanguage data.

3. Results and Discussions

3.1. The Development of CSL Learners' Dependency Distance

Dependency distance is an effective metric to measure syntactic complexity, and its computation based on syntactically tagged corpora is an effective and easy method (Hiranuma, 1998). Previous studies (Li & Yan, 2020; Ouyang & Jiang, 2018) show that dependency distance can also be used to measure the development of syntactic complexity in interlanguage.

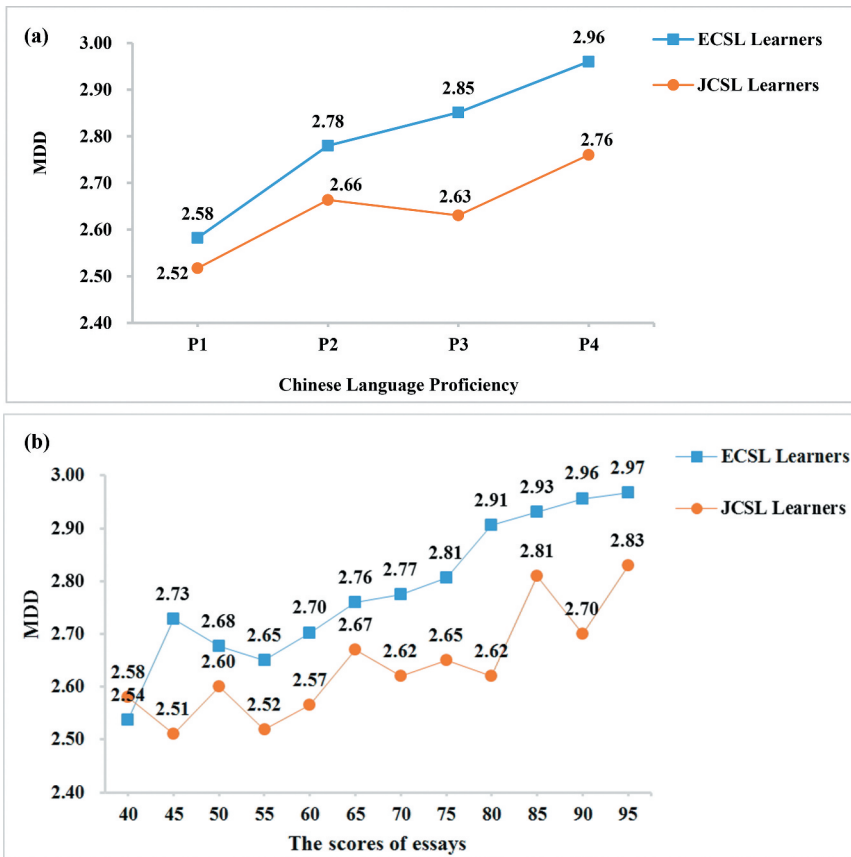


Figure 2. MDDs of Chinese interlanguage at each level of ECSL and JCSL learners. **Figure 2(a)** shows the relationship between four Chinese proficiency levels and dependency distance; **Figure 2(b)** displays the relation between 12 Chinese proficiency levels (or 12 types of essay scores) and dependency distance. (Note. MDD is the mean MDD over all the subjects/writings of certain proficiency level/score levels.)

Figure 2(a) and **Figure 2(b)** show the development for mean dependency distances of ECSL learners and JCSL learners across 4 levels and 12 levels respectively. As shown in **Figure 2(a)**, the MDDs of Chinese interlanguage of both types of CSL learners increase with their language proficiency, which is consistent with previous studies. The MDD of ECSL learners from P1 to P4 continues to rise from 2.58 to 2.96. The regression equation of the four levels of proficiency is $y = 0.1204x + 2.4923$, $p = .023$, $R^2 = 0.9562$, $n = 4$. In statistics, a sample of four points for regression analysis is relatively small, while five points is a borderline situation: it is the minimum sample size required to reject the null hypothesis in a two-sided correlation test using

a significance level of 0.05 (Ferrer-i-Cancho & Hernández-Fernández, 2013). Therefore, we took the level of target language as the representative of the highest level as an ideal state, and put the corresponding dependency distance data into the regression equation. The reason for such a treatment of adding the fifth point is that, according to the interlanguage theory, second language acquisition is a dynamic process of continuous accumulation and improvement. L2 learner's interlanguage proficiency gradually approaches the target language level and infinitely approaches the target language level. In other words, the level of the target language is the most ideal state for the second language learners, and it can represent the highest ideal level of second language learners (Nemser, 1971; Selinker, 1972). The correlation analysis indicates that the MDDs of ECSL learners is positively correlated with their language proficiency, and the regression equation is $y = 0.1215x + 2.49$ ($p = .001$, $R^2 = 0.9779$, $n = 5$).

In order to further ensure the effectiveness of statistics, more data points were added for linear regression analysis, with the composition scores of each writer taken as the independent variable and the average dependency distance of its corresponding composition as the dependent variable. However, it turns out that the goodness-of-fit of the regression analysis for both types of learners is extremely low (R^2 of ECSL and JCSL are 0.107 and 0.048, respectively). Additionally, the Pearson correlation coefficient of the composition scores and MDD for ECSL and JCSL learners is very low (ECSL: $r_s = 0.333$; JCSL: $r_s = 0.22$). Further analyses reveal that the data fails to meet the assumptions of linear regression in terms of independence and normality. To be more specific, the Durbin-Watson test shows that the D-W value for JCSL is only 0.976; and the residuals of regression line for both ECSL and JCSL learners does not conform to the normal distribution.

Therefore, we transferred to a new analysis where the mean dependency distance of each score was taken as the dependent variable and the essay scores of 12 types (as noted in the Section of *Material*) as the independent variable. According to the new classification, each of the 12 types of scores was deemed as a level of language proficiency. In this way, there are 12 levels, which allows the previous classification made of 4 levels to be further refined. In other words, the two classifications with different granularities are complementary. We first calculated the mean dependency distance of each composition at each score level ($n = 12$), and then the mean dependency distances of 12 score levels were figured out (The average MDD at a certain proficiency level is the result of dividing the sum of MDDs of all the essays at the same level by the number of compositions at this level). In this way, a more detailed account is given based on more data points while yielding the goodness-of-fit that is required for by the regression analysis. As shown in

Figure 2(b), results show that the MDD of ECSL learners is positively correlated with the essay scores. The regression equation is $y = 0.0356x + 2.5513$ ($p < .001$, $R^2 = 0.9002$, $n = 12$).

The MDDs of JCSL learners undergo a growth from 2.52 to 2.76 with their increasing second language proficiency, albeit there is no significant ($p = .098$) correlation between the MDD and 4 states of language proficiency in Figure 2(a). Subsequently, we put the MDD of TL into the regression equation as we have done in the analysis of MDDs of ECSL, it is apparent that although JCSL learners' MDDs decrease at the stage of P2 to P3, JCSL learners' MDD is still proportional to their language proficiency, and the regression equation is $y = 0.1262x + 2.356$ ($p = .038$, $R^2 = 0.8072$, $n = 5$). When the number of ranges of test scores increases to 12, as displayed in Figure 2(b), a significant difference can also be found in the correlation between the mean dependency distance and essay scores, and the corresponding regression equation is $y = 0.0232x + 2.4887$ ($p = 0.001$, $R^2 = 0.6837$, $n = 12$).

Subsequently, we conduct two ANOVA tests to observe whether there are significant differences between the two types of CSL learners at different levels ($n = 4$). We took the dependency distance in each raw dataset as the dependent variable and the language proficiency as the independent variable. According to one-way ANOVA test, there are significant differences ($ps < .001$) between different levels for each group of Chinese learners, namely, ECSL learners ($F(3, 75,496) = 53.492$) and JCSL learners ($F(3, 74,703) = 26.243$). A post-hoc analysis using Tukey-HSD shows that there are no significant differences ($p = .122$) in ECSL learners' MDDs between P2 ($M = 2.58$, $SD = 0.031$) and P3 ($M = 2.84$, $SD = 0.031$), but there are significant differences between all pairs ($ps < .05$). In terms of the MDD of JCSL learners, there is not significantly different ($ps = .675$) between P2 ($M = 2.66$, $SD = 0.0279$) and P3 ($M = 2.63$, $SD = 0.0279$), but significant differences are found between other pairs ($ps < .05$). The results show that only at the stage during P2-P3, MDDs of the two types of CSL learners do not change significantly, but in all the other pairs. In addition, we examined the MDDs of ECSL and JCSL learners by t-tests, and found that there are significant differences between ECSL and JCSL learners at all levels ($n = 4$), P1 ($t = 2.465$, $df = 36,852.108$), P2 ($t = 4.134$, $df = 37,482.209$), P3 ($t = 7.404$, $df = 37,006.374$) and P4 ($t = 6.305$, $df = 37,493.789$) show significant differences in the value of MDDs between the two types of CSL learners ($ps < .05$).

The mean of Chinese MDD we have calculated is 3.10, which is within the range of 2.02 ~ 3.66. It is similar to the results reported by Liu, Zhao et al. (2009). From the development trend of MDDs, both ECSL learners and JCSL learners show significant changes with the improvement of Chinese proficiency, which is in line with similar findings in studies on English as an interlanguage by Ouyang and Jiang (2018) and Li and Yan (2020). This

implies that MDD has some universal applicability as an effective metric to measure the syntactic development of interlanguage. Here, we give a positive answer to Question 1.

Specifically, the MDDs of ECSL learners are significantly higher than that of JCSL learners at all levels. As shown in [Figure 2](#), with the improvement of the second language proficiency, the differences of MDD between the two types of CSL learners grow larger. As for the process of second language acquisition, the development of ECSL learners' MDD is more significant and stable than that of JCSL learners. According to the data of advanced learning level (P4), ECSL learners' MDD is only 0.14 lower than TL (3.10), while JCSL learners' MDD is 0.34 lower than TL. We suggest that native language backgrounds of CSL learners may have a specific effect on the developmental trends of MDD and the specific values of MDD between two types of learners.

To begin with, although there is a statistical difference between the two types of L2 learners at the initial learning level (P1), the value of significance is .014 which is merely at a significance level of 0.05, while differences in other three levels are much larger than that of P1 ($ps < .001$). This may be due to the fact that second language learners are still at the stage of second language knowledge input at the initial, and they have not yet connected and reconstructed their native language knowledge with the new second language knowledge (VanPatten, 1996). With the improvement of L2 proficiency, learners of different native language backgrounds start to integrate their native language systems with their second language knowledge, and their interlanguage systems have undergone different changes. Additionally, from the perspective of linguistic typology, Chinese and English are both SVO languages, while Japanese is SOV language (Dryer, 2013). So, in terms of overall trends, L2 proficiency of ECSL learners whose native language (English) is more similar to the target language in linguistic typology developed more smoothly than that of JCSL learners and ECSL learners' MDD is closer to that of native Chinese learners at the final. Eventually, Liu (2008) shows that English MDD (2.54) is higher than that of Japanese (1.81), and dependency distance is closely related to the language's syntactic structure (Dryer, 1992; Liu, 2008; Temperley, 2008). Previous studies have shown that English and Japanese differ greatly in syntactic features (Fukui, 1988; Jeong et al., 2007; Shibatani, 1990). For example, English is the language that is dominated by noun-centred structures, which are not allowed to be left out (Halliday, 2004). Thus, the structure of sentences in English is long and complex. A large number of subordinate structures, conjunctions and relational words connect the various components to form different sentence patterns, and English is becoming analytical with its quite limited inflectional variations (Liu & Xu, 2012). While Japanese is rich in case markers instead of function words, its embedded structures are fewer than those in English (Nida, 1982; Oya, 2013). Consequently, these differences result in a higher

English MDD. Different grammatical features in language limit the cognitive mechanisms involved in sentence comprehension and production (Bates & MacWhinney, 1989). From the perspective of native language transfer, L2 learners may be influenced by the cognitive mechanisms of their native languages and thus transfer the features of their native languages to second language learning (Ramat, 2003). Therefore, there are differences with respect to the specific values of MDD between two types of learners. Specifically, ECSL learners' MDD is consistently higher than JCSL learners' across the four levels.

3.2. Probability Distribution of Dependency Distance

Psychological evidences show that there is a limit to the ability of human working memory (Miller, 1956). Human brains cannot store all of sentence units at a time during language processing. Instead, each unit is progressively combined for extraction and processing (Hawkins, 2004). Human languages follow the principle of least effort (Zipf, 1949). There is a general tendency to minimize dependency distance (Dependency Distance Minimization, DDM) in human language (Ferrer-i-Cancho, 2004; Futrell et al., 2015; Liu, 2008). The dependency distance between the governor and the dependent in a sentence cannot be extended indefinitely. This need for 'less effort' leads to the power-law probability distribution of dependency distance, which can fit the Zipf-Alekseev distribution (Jiang & Liu, 2015).

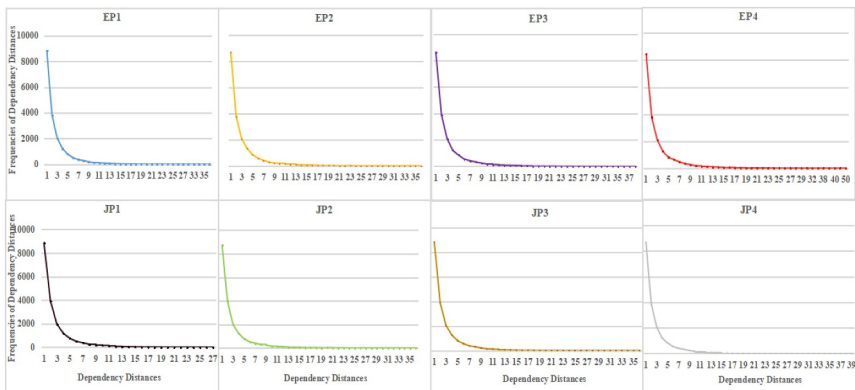


Figure 3. Probability distributions of dependency distances of different levels of CSL learners.

Table 3. Fitting the right truncated modified Zipf-Alekseev model to the dependency distances of different levels of ECSL and JCSL learners.

L2 proficiency	Parameters			a	χ^2	$P(\chi^2)$	DF	C	R^2
	a	b	n						
EP1	0.3451	0.5813	19	0.477	25.87	0.0269	14	0.0014	0.9997
EP2	0.3001	0.5462	20	0.4607	30.71	0.0096	15	0.0016	0.9997
EP3	0.6608	0.4282	24	0.4544	50.13	0.0001	19	0.0026	0.9997
EP4	0.7625	0.3701	22	0.4449	37.79	0.0026	17	0.002	0.9999
JP1	0.5774	0.5156	18	0.4805	53.01	0	13	0.0029	0.9996
JP2	0.5311	0.5046	20	0.4666	60.95	0	15	0.0033	0.9994
JP3	0.6832	0.4578	19	0.4701	31.28	0.0051	14	0.0017	0.9999
JP4	0.7028	0.4207	20	0.4618	38.1	0.0009	15	0.002	0.9998
TL	0.8480	0.3262	27.5	0.4753	57.035	0.0001	22.5	0.00295	0.9997

Note. n represents the types of different sentence lengths, the value of n for TL is the mean of two contrastive treebanks of native Chinese; χ^2 is the Chi-square value; $P(\chi^2)$ represents the Chi-square probability; DF indicates the degree of freedom; the coefficient of variation is indicated by C ; R^2 represents the goodness-of-fit.

Figure 3 presents the probability distributions of ECSL and JCSL learners' dependency distances from P1~ P4. It is obvious from the following figures that all the eight lines have the long tail effect of power-law curve, which indicates that the dependency distance of CSL learners also conforms to the above language rule.

Altmann-Fitter (2013) is used to fit the data of probability distribution of dependency distance of two types of learners with different native languages to the right truncated modified Zipf-Alekseev distribution and we get the data in Table 3. The determination coefficient R^2 is generally considered to be one of the criteria for goodness-of-fit when the data set is large. It can be seen that the eight sets of datum of learners with two different L1 backgrounds fit the right truncated modified Zipf-Alekseev well, and the values of all groups' determination coefficient are above 0.99. Although two types of learners' MDD cannot reach the TL's level across four levels, their probability distributions of dependency distance follow the same law as that of native Chinese speakers. Our results make a response to Question 2. In accordance with the previous research results, the fitting results of the dependency distance distribution in Chinese interlanguage are similar to that of natural languages. They both follow the principle of least effort to construct their respective language systems.

3.3. Similarities and Differences in the Fitting Parameters between Two Types of CSL Learners

Ouyang and Jiang (2018) and Li and Yan (2020) have examined English interlanguage and found that the parameters a and b in the Zipf-Alekseev model are correlated with the development of English

interlanguage proficiency. It is further believed that the two parameters can also be used as metrics to measure L2 proficiency. We have shown that probability distribution of interlanguage dependency distance in Chinese can also well fit the right truncated modified Zipf-Alekseev distribution. By investigating the distribution of word-length in dozens of languages, Popescu et al. (2014) have found that fitting parameter values in terms of language unit length is likely to vary in different languages. The systems of Chinese, English and Japanese (one of them as target language, and the other two as interlanguage) are different. Accordingly, the fitting parameter values in terms of interlanguage unit length are supposed to be distinct. Do the values of a and b of the right truncated modified Zipf-Alekseev distribution in Chinese interlanguage

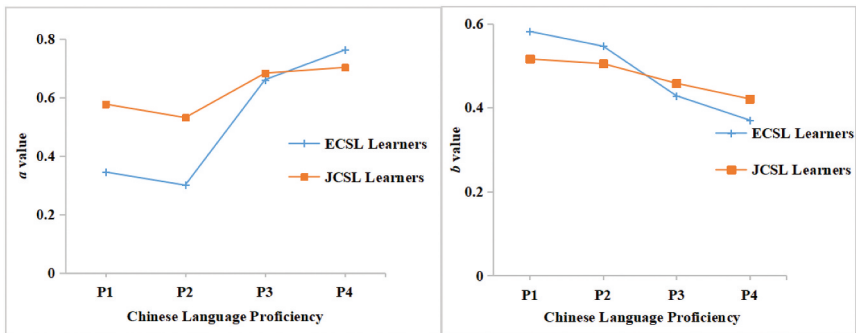


Figure 4. The development of parameters a (left) and b (right) of the right truncated modified Zipf-Alekseev, fitting the dependency distances of different levels of CSL Learners.

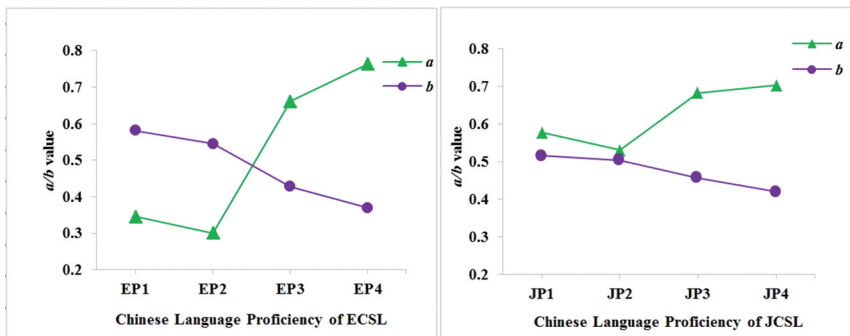


Figure 5. The correlations between the parameters (a , b) of CSL learners with different native backgrounds.

also correlate with language proficiency? We further answer whether the fitting parameters a and b can be used as the universal metric to measure L2 proficiency, which is brought up in Question 3.

According to Figure 4, we can see that the values of parameters a and b of the right truncated modified Zipf-Alekseev distribution change with the improvement of Chinese proficiency, which is true of both ECSL and JCSL learners. As shown in Figure 5, it can be clearly seen that the parameters a and b of the probability distribution of CSL learners with different native language backgrounds have opposite overall changing trends with increasing Chinese proficiency. Firstly, we conducted a linear regression analysis (same as above processing of dependency distance datum, we took the values of parameters a and b of the right truncated modified Zipf-Alekseev distribution in Chinese as the representative of the highest level of L2 learners ideally). The correlation coefficient between the value of ECSL learners' a , b and language proficiency was 0.938 and -0.984 respectively ($ps < 0.05$). The correlation coefficient between the value of ECSL learners' a , b and language proficiency was 0.914 and -0.956 , respectively ($ps < .05$). There is a significant positive correlation between ECSL learners' fitting parameter a and Chinese proficiency, the regression equation is $y = 0.1468x + 0.1429$, $p = .018$, $R^2 = 0.8794$, $n = 5$; the fitting parameter b has a significant negative correlation with Chinese proficiency, and the regression equation is $y = -0.0686x + 0.6563$, $p = .002$, $R^2 = 0.9676$, $n = 5$. The regularity holds true for JCSL learners' fitting parameters. The regression equation of fitting parameter a is $y = 0.0713x + 0.4547$, $p = .03$, $R^2 = 0.836$, $n = 5$; the regression equation of fitting parameter b is $y = -0.0463x + 0.5838$, $p = .011$, $R^2 = 0.9147$, $n = 5$. The value of a and b of both types of CSL learners do not reach the TL's level at P4. In addition, there are differences in the development of the parameters a and b of the probability distribution of dependency distance between the ECSL and JCSL learners. Although the fitting parameter a of JCSL learners is larger than that of ECSL learners at the early stages, it is the same at P3 stages, with the emphasis on the fact that the parameter value of ECSL learners is higher in the final state. The development of fitting parameter b of the ECSL and JCSL learners follows similar rules. In general, the values of fitting parameters of ECSL learners are more significant than those of JCSL learners. The fitting parameter values of ECSL learners whose language typological features are more similar to the target language are closer to those of Chinese native speakers than those of JCSL learners at the advanced level (P4).

In conclusion, as CSL learners' language proficiency increases, our results are consistent with the findings from Ouyang and Jiang (2018) and Li and Yan (2020). We also find that the fitting parameters a and b of probability distribution of Chinese interlanguage's dependency distance increase and decrease, respectively; meanwhile, they are all approaching the target language level gradually. It is proved that the fitting parameters a and b of the right truncated

modified Zipf-Alekseev distribution as a metric to measure L2 proficiency development, which is applicable not only to English as an interlanguage, but also to the less inflectional Chinese as an interlanguage. Furthermore, we find that to a great extent, the fitting parameters a and b of the probability distribution may be influenced by the L2 learner's native language backgrounds. For example, at the final state, the value of a in ECSL learners' interlanguage is closer to that of the Chinese target language than that of JCSL learners. The value of a in ECSL learners' interlanguage is only 0.085 lower than that of the target language, while that of JCSL learners is nearly 0.15 lower than that of the target language.

4. Conclusions and Implications

In this paper, we have investigated the development of the dependency distance of Chinese interlanguage system with different L1 backgrounds. Materials from the HSK composition corpus from learners of four L2 levels with English or Japanese native language backgrounds were syntactically annotated and analysed, compared with the target language (Chinese) data. Furthermore, the probability distribution of dependency distance was studied. The results show that although CSL learners' native language backgrounds are different, their MDDs vary significantly with the development of their Chinese proficiency, and probability distribution of MDD can also successfully fit the right truncated modified Zipf-Alekseev distribution. We further found that the parameters a and b of right truncated modified Zipf-Alekseev distribution can also reflect the development of proficiency in Chinese interlanguage. This paper demonstrated the universal applicability of dependency distance and the parameters of probability distribution of MDD as metrics to measure the development of language proficiency of interlanguages. We have examined the influence of L1 backgrounds on the development of second language learners' MDD and probability distribution of dependency distance, and further drawn the following conclusions.

Specifically, there are many similarities between the DDMs development of ECSL and JCSL learners. Regarding the MDDs of two types CSL learners, although there is no statistical difference between P2 and P3, with the improvement of language proficiency, it is obvious that their MDDs gradually approach the target language. However, at the advanced learning level, they both fail to reach the target language level. The process of second language acquisition is complex and dynamic, and its temporary stagnation actually represents an infinite potential (Larsen-Freeman, 1997). In addition to the commonality, the developmental features of the two types of CSL learners from different native language backgrounds are also distinctive. Contrastive linguistic research shows that English sentences are flatter when viewed in syntactic trees and more embedded than their Japanese translation. Reflected in the dependency distance, English MDD

is higher than that of Japanese. Influenced by their own native languages, the MDD of ECSL learners from P1 to P4 is higher than that of JCSL learners. In addition, English is more similar to Chinese than Japanese is. Because of the language distance, the development of ECSL learners' MDD is more stable, and the MDD of ECSL learners is closer to that of Chinese native speakers at the advanced learning level. With respect to the parameters of the right truncated modified Zipf-Alekseev distribution of dependency distance, the values of a and b of ECSL learners and JCSL learners differ from the target language Chinese at the beginning learning level. The gap between the interlanguage and the target language has gradually narrowed, but none of them has reached the target language level. Same as the law of the development of MDD, in terms of the values of a or b , the change of ECSL learners is more significant and stable than that of JCSL learners. At the advanced learning level, the values of a and b in ECSL learners' interlanguage are also closer to those of TL than JCSL learners'. In conclusion, we argue that the second language acquisition of L2 learners with different native language backgrounds is a constructive process with the target language as the reference. L2 learners are also affected by the native language. There are differences in terms of the process and the final state of L2 acquisition.

In this paper, taking advantage of the method commonly used in quantitative linguistics, we have demonstrated the universal applicability of dependency distance and its probability distribution as metrics to measure interlanguage proficiency. In addition, from the perspective of linguistic typology, we have further explored the influence of the native language on the development of dependency distance of interlanguage produced by CSL learners from different native language backgrounds. Once again, this study has demonstrated the feasibility of the quantitative linguistics metrics applied in the field of second language acquisition. However, the present study does not investigate the similarities and differences in the development of the dependency distance between native Chinese and second Chinese writings on the same topic after similar learning time, which may have influenced the results and are needed to control in future studies. Additionally, we will adopt the fine-grained linear regression analyses (such as linear mixed-effect models) in the future research and pay more attention to the development of dependency distance or syntactic complexity of low-level L2 learners. The conclusions reached in this paper still need to be further verified by the following studies on more languages.

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ORCID

Yanni Lin  <http://orcid.org/0000-0001-5621-2633>

References

- Altmann-Fitter. (2013). *Altmann-Fitter user guide. The 3.1 version*. Retrieved May 4, 2020, from <https://www.ram-verlag.eu/wp-content/uploads/2013/08/Fitter-User-Guide.pdf>
- Altmann, G. (1980). Prolegomena to Menzerath's law. In R. Grotjahn (Ed.), *Glottometrika* (Vol. 2, pp. 1–10). Brockmeyer.
- Andreev, V. (2020). Fitting the distribution of the syllabic types in different positions of verse. *Glottometrics*, 49, 87–97. <https://ram-verlag.org/382-fitting-the-distribution-of-the-syllabic-typesin-different-positions-of-verse>
- Bates, E., & MacWhinney, B. (1989). Functionalism and the competition model. *The Crosslinguistic Study of Sentence Processing*, 3, 73–112. https://www.academia.edu/download/44487135/Functionalism_and_the_Competition_Model20160406-28539-9h7uea.pdf
- Beöthy, E., & Altmann, G. (1982). Das Piotrowski-Gesetz und der Lehnwortschatz. *Zeitschrift für Sprachwissenschaft*, 1(2), 171–178. <https://doi.org/10.1515/zfsw.1982.1.2.171>
- Bi, P., & Jiang, J. (2020). Syntactic complexity in assessing young adolescent EFL learners' writings: Syntactic elaboration and diversity. *System*, 91, 102248. <https://doi.org/10.1016/j.system.2020.102248>
- Brown, H. D. (1994). *Principles of language learning and teaching* (3rd ed.). Prentice Hall Regents.
- Che, W., Li, Z., & Liu, T. (2010). LTP: A Chinese language technology platform. In *Proceedings of the Coling Demonstrations* (pp. 13–16). Beijing, China. <http://ltp.ai/>
- Chen, H., & Xu, H. (2019). Quantitative linguistics approach to interlanguage development: A study based on the Guangwai-Lancaster Chinese Learner Corpus. *Lingua*, 230, 102736. <https://doi.org/10.1016/j.lingua.2019.102736>

- Corder, S. (1967). The significance of learner's errors. *IRAL - International Review of Applied Linguistics in Language Teaching*, 5(1–4), 161–170. <https://doi.org/10.1515/iral.1967.5.1-4.161>
- Dryer, M. S. (1992). The Greenbergian word order correlations. *Language*, 68(1), 81–138. <https://doi.org/10.1533/lan.1992.0028>
- Dryer, M. S. (2013). Order of subject, object and verb. In M. S. Dryer, and M. Haspelmath (Eds.), *The world atlas of language structures online*. Max Planck Institute for Evolutionary Anthropology. Retrieved 25 October 2021 from <http://wals.info/chapter/81>
- Eppler, E. (2011). The dependency distance hypothesis for bilingual code-switching. *Proceedings of dependency linguistics* (pp. 145–154). Barcelona. Retrieved August 4, 2020, from <http://depling.org/proceedingsDepling2011/papers/eppler.pdf>
- Eppler, E. (2010). *Emigranto: The syntax of German-English code-switching*. Wilhelm Braumüller.
- Ferrer-i-Cancho, R., Gómez-Rodríguez, C., Esteban, J. L., & Alemany-Puig, L. (2020). *The optimality of syntactic dependency distances*. ArXiv. [abs/2007.15342](https://arxiv.org/abs/2007.15342)
- Ferrer-i-Cancho, R., & Hernández-Fernández, A. (2013). The failure of the law of brevity in two new World primates. *Statistical Caveats. Glottotheory*, 4(1), 45–55. <https://doi.org/10.1524/glot.2013.0004>
- Ferrer-i-Cancho, R. (2004). Euclidean distance between syntactically linked words. *Physical Review E*, 70(5), 056135. <https://doi.org/10.1103/physreve.70.056135>
- Fukui, N. (1988). Deriving the differences between English and Japanese: A case study in parametric syntax. *English Linguistics*, 5, 249–270. <https://doi.org/10.9793/elsj1984.5.249>
- Futrell, R., Mahowald, K., & Gibson, E. (2015). Large-scale evidence of dependency length minimization in 37 languages. *Proceedings of the National Academy of Sciences*, 112(33), 10336–10341. <https://doi.org/10.1073/pnas.1502134112>
- Gass, S. M., & Selinker, L. (1992). *Language transfer in language learning* (Revised ed.). John Benjamins.
- Gibson, E. (1998). Linguistic complexity: Locality of syntactic dependencies. *Cognition*, 68(1), 1–76. [https://doi.org/10.1016/S0010-0277\(98\)00034-1](https://doi.org/10.1016/S0010-0277(98)00034-1)
- Gibson, E. (2000). The dependency locality theory: A distance-based theory of linguistic complexity. In A. Marantz, Y. Miyashita, & W. O'Neil (Eds.), *Image, language, brain* (pp. 95126). 95126). The MIT Press.
- Gildea, D., & Temperley, D. (2010). Do grammars minimize dependency length? *Cognitive Science*, 34(2), 286–310. <https://doi.org/10.1111/j.1551-6709.2009.01073.x>
- Halliday, M. (2004). *The language of science*. Continuum.
- Hawkins, J. A. (2004). *Efficiency and complexity in grammars*. Oxford University Press.
- Herdan, G. (1964). *Quantitative linguistics*. Butterworth.
- Heringer, H. J., Strecker, B., & Wimmer, R. (1980). *Syntax: Fragen-Lösungen-Alternativen*. München. Wilhelm Fink Verlag.
- Hiranuma, S. (1998). Measuring the processing load of Japanese words. *UCL Working Papers in Linguistics*, 10, 1–28. <https://www.phon.ucl.ac.uk/home/PUB/WPL/98papers/hiranuma.pdf>
- Hiranuma, S. (1999). Syntactic difficulty in English and Japanese: A textual study. *UCL Working Papers in Linguistics*, 11, 309–322.
- Hřebíček, L. (2002). Zipf's law and text. *Glottometrics*, 3, 27–38. <https://d-nb.info/1165200732/34#page=31>

- Hřebíček, L. (1996). Word associations and text. In P. Schmidt (Ed.), *Glottometrika* (Vol. 15, pp. 12–17). Wissenschaftlicher Verlag Trier.
- Hudson, R. (1995). Measuring syntactic difficulty [Unpublished paper]. Retrieved August 13, 2020 from <https://dickhudson.com/wp-content/uploads/2013/07/Difficulty.pdf>
- Hudson, R. (2007). *Language networks: The new word grammar*. Oxford University Press.
- Jeong, H., Sugiura, M., Sassa, Y., Haji, T., Usui, N., Taira, M., Horie, K., Sato, S., & Kawashima, R. (2007). Effect of syntactic similarity on cortical activation during second language processing: A comparison of English and Japanese among native Korean Trilinguals. *Human Brain Mapping*, 28(3), 194–204. <https://doi.org/10.1002/hbm.20269>
- Jiang, J., & Liu, H. (2015). The effects of sentence length on dependency distance, dependency direction and the implications—based on a parallel English–Chinese dependency treebank. *Language Sciences*, 50, 93–104. <https://doi.org/10.1016/j.langsci.2015.04.002>
- Jiang, J., & Ouyang, J. (2017). Dependency distance: A new perspective on the syntactic development in second language acquisition: Comment on “Dependency distance: A new perspective on syntactic patterns in natural language” by Haitao Liu et al. *Physics of Life Reviews*, 21, 209–210. <https://doi.org/10.1016/j.plrev.2017.06.018>
- Jing, J., & Liu, H. (2015). Mean hierarchical distance augmenting mean dependency distance. *Proceedings of the Third International Conference on Dependency Linguistics*, 161–170. <https://aclanthology.org/W15-2119.pdf>
- Köhler, R., Altmann, G., & Piotrowski, R. G. (2005). *Quantitative linguistics-An international handbook*. Mouton de Gruyter.
- Komori, S., Sugiura, M., & Li, W. (2019). Examining MDD and MHD as syntactic complexity measures with intermediate Japanese learner corpus data. *Proceedings of the Fifth International Conference on Dependency Linguistics (Depling, SyntaxFest 2019)*, 130–135. <https://aclanthology.org/W19-7715.pdf>
- Larsen-Freeman, D. (1997). Chaos/complexity science and second language acquisition. *Applied Linguistics*, 18(2), 141–165. <https://doi.org/10.1093/applin/18.2.141>
- Li, W., & Yan, J. (2020). Probability distribution of dependency distance based on a treebank of Japanese EFL learners’ interlanguage. *Journal of Quantitative Linguistics*, 2, 1–15. <https://doi.org/10.1080/09296174.2020.1754611>
- Liu, H., Hudson, R., & Feng, Z. (2009). Using a Chinese treebank to measure dependency distance. *Corpus Linguistics and Linguistic Theory*, 5(2), 161–174. <https://doi.org/10.1515/CLLT.2009.007>
- Liu, H., & Xu, C. (2012). Quantitative typological analysis of romance languages. *Poznań Studies in Contemporary Linguistics*, 48(4), 597–625. <https://doi.org/10.1515/psicl-2012-0027>
- Liu, H., Zhao, Y., & Li, W. (2009). Chinese syntactic and typological properties based on dependency syntactic treebanks. *Poznań Studies in Contemporary Linguistics*, 45(4), 509–523. <https://doi.org/10.2478/v10010-009-0025-3>
- Liu, H. (2007). Probability distribution of dependency distance. *Glottometrics*, 15, 1–12. <https://www.ram-verlag.eu/wp-content/uploads/2018/08/glo15abstract.pdf>
- Liu, H. (2008). Dependency distance as a metric of language comprehension difficulty. *Journal of Cognitive Science*, 9(2), 159–191. <https://doi.org/10.17791/jcs.2008.9.2.159>

- Liu, H. (2009). *Dependency grammar: From theory to practice*. Science Press.
- Liu, H. (2010). Dependency direction as a means of word-order typology: A method based on dependency treebanks. *Lingua*, 120(6), 1567–1578. <https://doi.org/10.1016/j.lingua.2009.10.001>
- Liu, H. (2017). *An introduction to quantitative linguistics*. The Commercial Press.
- Lu, Q., & Liu, H. (2016). Does dependency distance distribute regularly? *Journal of Zhe Jiang University (Humanities and Social Sciences Online Edition)*, 4, 1–14. <https://kns.cnki.net/kcms/detail/detail.aspx?FileName=ZJDX201604006&DbName=CJFQ2016>
- Lu, X. (2011). A corpus-based evaluation of syntactic complexity measures as indices of college-level ESL writers' language development. *TESOL Quarterly*, 45(1), 36–62. <https://doi.org/10.5054/tq.2011.240859>
- Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63(2), 81–97. <https://doi.org/10.1037/h0043158>
- Morrill, G. (2000). Incremental processing and acceptability. *Computational Linguistics*, 26(3), 319–338. <https://doi.org/10.1162/089120100561728>
- Nemser, W. (1971). Approximative systems of Foreign language learners. *IRAL-Interlanguage Review of Applied Linguistics in Language Learning*, 9(2), 115–124. <https://doi.org/10.1515/iral.1971.9.2.115>
- Nida, E. A. (1982). *Translating meaning*. English Language Institute.
- Odlin, T. (1989). *Language transfer: Cross-linguistic influence in language learning*. Cambridge University Press. <http://dx.doi.org/10.1017/CBO9781139524537>
- Ortega, L. (2003). Syntactic complexity measures and their relationship to L2 proficiency: A research synthesis of college-level L2 writing. *Applied Linguistics*, 24(4), 492–518. <https://doi.org/10.1093/applin/24.4.492>
- Ouyang, J., & Jiang, J. (2018). Can the probability distribution of dependency distance measure language proficiency of second language learners? *Journal of Quantitative Linguistics*, 25(4), 295–313. <https://doi.org/10.1080/09296174.2017.1373991>
- Oya, M. (2013). Syntactic dependency structures of English and Japanese. *Meiji Journal of Humanities* 9, 151–164. <https://core.ac.uk/download/pdf/228932242.pdf>
- Popescu, I. I., Best, K. H., & Altmann, G. (2014). *Unified modeling of length in language (= studies in quantitative linguistics 16)*. RAM-Verlag. 978-3942303-26-2.
- Ramat, A. G. (2003). *Typology and second language acquisition. The encyclopedia of applied linguistics*. American Cancer Society.
- Richards, J. C., Plott, J., & Platt, H. (1996). *Dictionary of language teaching and applied linguistics*. Longman.
- Selinker, L. (1972). Interlanguage. *IRAL - International Review of Applied Linguistics in Language Teaching*, 10(3), 219–231. <https://doi.org/10.1515/iral.1972.10.1-4.209>
- Shibatani, M. (1990). *The languages of Japan*. Cambridge University Press.
- Strauss, U., & Altmann, G. (2006). *Diversification - laws in quantitative linguistics*. Retrieved April 8, 2021, from http://www.uni-trier.de/uni/fb2/ldv/lql_wiki/index.php/Diversification
- Strauss, U., Grzybek, P., & Altmann, G. (2006). Word length and word frequency. In P. Grzybek (Ed.), *Contributions to the science of text and language: Word length studies and related issues* (pp. 277294). 277294). Springer.
- Temperley, D. (2007). Minimization of dependency length in written English. *Cognition*, 105(2), 300–333. <https://doi.org/10.1016/j.cognition.2006.09.011>

- Temperley, D. (2008). Dependency-length minimization in natural and artificial languages. *Journal of Quantitative Linguistics*, 15(3), 256–282. <https://doi.org/10.1080/09296170802159512>
- Tesnière, L. (1959). *Eléments de la syntaxe structurale*. Klincksieck.
- VanPatten, B. (1996). *Input processing and grammar instruction: Theory and research*. Ablex.
- Wolfe-Quintero, K., Inagaki, S., & Kim, H. Y. (1998). *Second language development in writing: Measures of fluency, accuracy, & complexity*. Second Language Teaching and Curriculum Center, University of Hawaii Press.
- Yan, H., & Li, Y. (2019). Beyond length: Investigating dependency distance across L2 modalities and proficiency levels. *Open Linguistics*, 5(1), 601–614. <https://doi.org/10.1515/opli-2019-0033>
- Zipf, K. G. (1936). *The psychobiology of language*. Routledge.
- Zipf, K. G. (1949). *Human behavior and the principle of least effort: An introduction to human ecology*. Addison-Wesley.